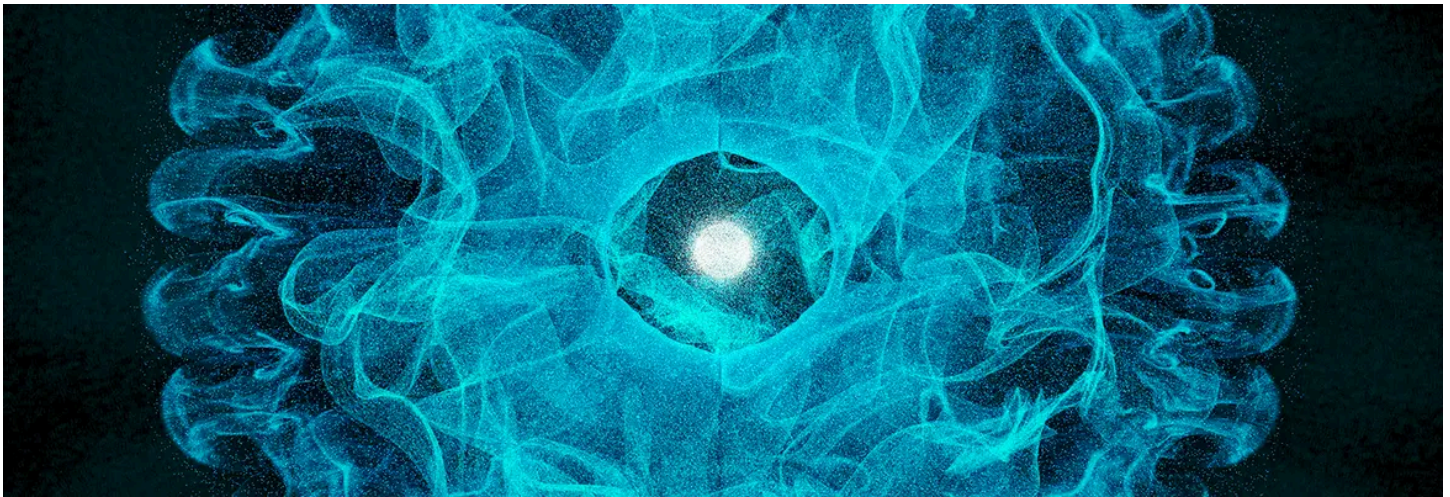


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Penrose's theory of consciousness supported by new quantum evidence

New neuroscientific study showcases warm quantum states



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Mike Wiest | Professor of Neuroscience at Wellesley College.

1,969 words

Read time: approx. 10 mins

*Roger Penrose argued that human thought does things no computer can, so the mind must make use of non-computable quantum mechanics. The idea remains fringe. Most scientists think that in warm, wet environments like the brain, quantum states collapse into classical states far faster than neurons fire, and so cognition must be governed by classical physics. But neuroscientist **Mike Wiest** argues that new evidence challenges this assumption: stable quantum states are now being observed at far higher temperatures than previously thought possible. And consciousness, he suggests, has structural features that make quantum mechanics a natural place to look for an explanation, so critics should stop accusing Penrose of simply piling one mystery on top of another. The time has come to give this neglected proposal another look.*

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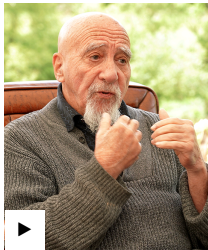
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I. Introduction: quantum consciousness

Quantum physics is generally understood as the physics of the very small. The idea that it might be relevant to understanding how the brain generates our conscious experience has been considered laughable since it was proposed by Nobel Laureate Roger Penrose and others in the 1990s. A popular way to ridicule its proponents, to this day, has been to say they are jamming two unrelated mysterious subjects together, namely quantum physics and consciousness, and declaring them both solved. As I will explain below, this is far from the truth, because there is a clear and decisive conceptual motivation for pursuing a quantum model of biological consciousness.

But first I must address the main *legitimate* reason for skepticism regarding the plausibility of this “quantum consciousness” hypothesis, which is that the brain is too *warm* to sustain a delicate quantum process long enough for it to be useful. To be sure, processes at the atomic level are governed by quantum physics in all matter including brains—this is uncontroversial. But for many practical purposes, these atomic-level quantum effects represent tiny random fluctuations that tend to average out at larger scales, such that classical physical descriptions are adequate.



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To harness quantum physics for computational purposes (beyond as a mere source of randomness) requires *coordinating* the quantum activities of very large numbers of atoms. Even though Penrose argued that consciousness requires non-computable physics (see Section III below), he does not deny that neurons perform computations. In other words, Penrose argues that conscious understanding is not reducible *solely* to computation. His theory still requires biological structures to “orchestrate” many molecular activities so that the non-computable, conscious part of the process is meaningful and useful to an organism.

Since higher temperatures reflect more violent random jostling at the atomic level, our attempts so far to engineer quantum technology like quantum computers or superconductors have usually required ultra-cold temperatures far below zero. This is to prevent the random molecular movements at warmer temperatures from destroying the quantum coherence—that is, coordination—among parts of the system. This is the source of the intuition that the brain is too warm to sustain *useful* quantum coherence. This [temperature objection](#) was [never conclusive](#), but in recent years experimental evidence has accumulated from multiple sources to [support the physical plausibility of the quantum consciousness hypothesis](#).



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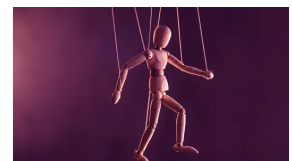
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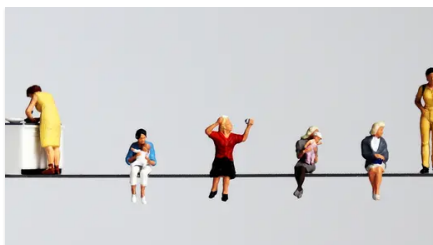


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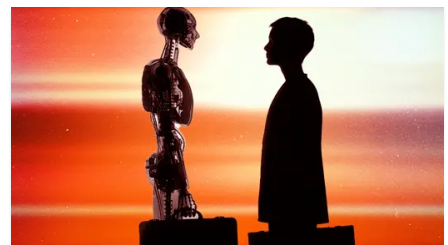
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Robert Durkacz

6 September 2026

It is implausible. Human brains work using the same mechanism that animal brains do. We obviously do not need any far-fetched hypotheses about quantum computing to account for animal behaviour.

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"efficacious unity of consciousness"...

I am not a scientist, but a retired teacher, who studied Adwaita Vedanta in India for a year. If all of your discussion is about "hosting" conscious, then perhaps. But as far as creating or uniting consciousness, that is not possible, for consciousness is One, exists a prior to the creation, it creating and then supporting creation in every way.

Consciousness is Sat, existence, Chit, intelligent, Ananda, unending, looking out our eyes from birth, and we using its knowledge speaking to us through our heart and right brain, receiving from the quantum field, the left brain to shape those inspirations into contributions for wellness of the entire world. Further: Tat Twam Asi, Sanskrit for: You Are That consciousness, not your limited human ego.

Reply 

Excellent article. It appears that the brain is far greater than any computer.

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